Weekly Quiz 1

Instructor Key

Sets, diagrams, and counting

20 minutes • 10 points • Show the essential steps.

1. Express "A or B, but not both" using unions, intersections, and complements. **[3 points]**

Solution

The event is the symmetric difference

$$(A \cap B^c) \cup (A^c \cap B),$$

equivalently $(A \cup B) \setminus (A \cap B)$.

2. A fair coin is tossed three times. Find the probability of exactly two heads. **[3 points]**

Solution

There are $2^3 = 8$ equally likely strings. Choose the two positions occupied by heads in $\binom{3}{2} = 3$ ways, so the probability is $3/8$.

3. In a group of 70 students, 38 study French, 27 study Spanish, and 12 study both. How many study neither language? **[4 points]**

Solution

Inclusion–exclusion gives

$$|F \cup S| = |F| + |S| - |F \cap S| = 38 + 27 - 12 = 53.$$

Therefore $70 - 53 = 17$ students study neither.

Weekly Quiz 2

Instructor Key

Probability spaces and events

20 minutes • 10 points • Show the essential steps.

1. Two fair dice are rolled. Find the probability that the maximum is 4. **[4 points]**

Solution

There are $4^2 = 16$ outcomes with both dice at most 4, and $3^2 = 9$ with both at most 3. Thus $16 - 9 = 7$ of the 36 outcomes have maximum 4, giving probability $7/36$.

2. Find the probability of at least one six in four independent rolls of a fair die. **[3 points]**

Solution

The complement is no six in all four rolls, which has probability $(5/6)^4$. Therefore

$$\mathbb{P}(\text{at least one six}) = 1 - \left(\frac{5}{6}\right)^4 = \frac{671}{1296}.$$

3. Derive $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ by splitting events into disjoint pieces. **[3 points]**

Solution

The union is the disjoint union $A \cup (B \setminus A)$, so $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. Also $B = (B \setminus A) \cup (A \cap B)$ disjointly, hence $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substitution gives the formula.

Weekly Quiz 3

Instructor Key

Conditional probability and Bayes' rule

20 minutes • 10 points • Show the essential steps.

1. An urn contains 3 red and 2 blue balls. Two are drawn without replacement. Find the probability that the second ball is red. **[3 points]**

Solution

By conditioning on the first draw,

$$\mathbb{P}(R_2) = \frac{3}{5}\frac{2}{4} + \frac{2}{5}\frac{3}{4} = \frac{6+6}{20} = \frac{3}{5}.$$

(Symmetry also says every draw position is red with probability $3/5$.)

2. A condition has prevalence 2%. A test has sensitivity 90% and specificity 95%. Find the probability that a person who tests positive has the condition. **[4 points]**

Solution

For a random person,

$$\mathbb{P}(D \cap +) = 0.02(0.90) = 0.018,$$

while the false-positive probability is $0.98(0.05) = 0.049$. Bayes' rule gives

$$\mathbb{P}(D \mid +) = \frac{0.018}{0.018 + 0.049} = \frac{18}{67} \approx 0.269.$$

3. Suppose $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.2$. Are A and B independent? Find $\mathbb{P}(A \mid B)$. **[3 points]**

Solution

Since $\mathbb{P}(A)\mathbb{P}(B) = 0.4(0.5) = 0.2 = \mathbb{P}(A \cap B)$, the events are independent. Also $\mathbb{P}(A \mid B) = 0.2/0.5 = 0.4$.

Weekly Quiz 4

Instructor Key

Random variables, PMFs, densities, and CDFs

20 minutes • 10 points • Show the essential steps.

1. A random variable X takes values $0, 1, 2$ with probabilities $0.2, 0.5, 0.3$. Write its CDF $F(x) = \mathbb{P}(X \le x)$ piecewise. **[4 points]**

Solution

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.2, & 0 \le x < 1, \\ 0.7, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

2. Let $f(x) = cx$ for $0 \le x \le 2$ and $f(x) = 0$ otherwise. Find c and $\mathbb{P}(X > 1)$. **[4 points]**

Solution

Normalization gives $1 = \int_0^2 cx\,dx = 2c$, so $c = 1/2$. Then

$$\mathbb{P}(X > 1) = \int_1^2 \frac{x}{2}\,dx = \frac{x^2}{4}\Big|_1^2 = \frac{3}{4}.$$

3. If X has a continuous distribution, what is $\mathbb{P}(X = 1)$? Why? **[2 points]**

Solution

It is 0: a single point has zero width, so its probability is $\int_1^1 f(x)\,dx = 0$.

Weekly Quiz 5

Instructor Key

Expectation, variance, and moments

20 minutes • 10 points • Show the essential steps.

1. For X with probabilities $0.2, 0.5, 0.3$ at $0, 1, 2$, compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$. **[4 points]**

Solution

$$\mathbb{E}[X] = 0(0.2) + 1(0.5) + 2(0.3) = 1.1,$$

and $\mathbb{E}[X^2] = 0 + 1(0.5) + 4(0.3) = 1.7$. Therefore $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 1.7 - 1.21 = 0.49$.

2. If $X \sim \mathrm{Bin}(20, 0.3)$, find its mean and variance. Explain briefly using indicators. **[3 points]**

Solution

Write $X = I_1 + \cdots + I_{20}$, where each I_i is Bernoulli(0.3). Then

$$\mathbb{E}[X] = 20(0.3) = 6, \qquad \mathrm{Var}(X) = 20(0.3)(0.7) = 4.2,$$

using independence for the variance sum.

3. Given $\mathbb{E}[X] = 4$ and $\mathbb{E}[Y] = -1$, find $\mathbb{E}[3X - 2Y + 5]$. Is independence needed? **[3 points]**

Solution

By linearity, $\mathbb{E}[3X - 2Y + 5] = 3(4) - 2(-1) + 5 = 19$. Independence is not needed for linearity of expectation.

Weekly Quiz 6

Instructor Key

Standard probability distributions

20 minutes • 10 points • Show the essential steps.

1. If $X \sim \text{Bin}(5, 0.4)$, find $\mathbb{P}(X = 2)$. **[3 points]**

Solution

$$\mathbb{P}(X = 2) = \binom{5}{2}(0.4)^2(0.6)^3 = 10(0.16)(0.216) = 0.3456.$$

2. A waiting time is exponential with mean 4 minutes. Find the probability that the wait exceeds 6 minutes. **[3 points]**

Solution

The rate is $\lambda = 1/4$ per minute. Therefore

$$\mathbb{P}(X > 6) = e^{-\lambda 6} = e^{-3/2} \approx 0.223.$$

3. Suppose $X \sim N(100, 15^2)$. Express $\mathbb{P}(X \le 130)$ using the standard normal CDF Φ, and evaluate using $\Phi(2) \approx 0.9772$. **[4 points]**

Solution

Standardizing,

$$\mathbb{P}(X \le 130) = \mathbb{P}\left(Z \le \frac{130 - 100}{15}\right) = \mathbb{P}(Z \le 2) = \Phi(2) \approx 0.9772.$$

Weekly Quiz 7

Instructor Key

Joint, marginal, and conditional distributions

20 minutes • 10 points • Show the essential steps.

For the next three questions, use the joint probability table

	$Y=0$	$Y=1$	$\mathbb{P}(X=x)$
$X=0$	0.10	0.20	
$X=1$	0.30	0.40	
$\mathbb{P}(Y=y)$			1

1. Find both marginal distributions. **[3 points]**

Solution

Row sums give $\mathbb{P}(X=0)=0.30$ and $\mathbb{P}(X=1)=0.70$. Column sums give $\mathbb{P}(Y=0)=0.40$ and $\mathbb{P}(Y=1)=0.60$.

2. Find $\mathbb{P}(X=1 \mid Y=0)$ and decide whether X and Y are independent. **[3 points]**

Solution

$\mathbb{P}(X=1 \mid Y=0)=0.30/0.40=0.75$. They are not independent: for example, $\mathbb{P}(X=1, Y=0)=0.30 \neq 0.70(0.40)=0.28$.

3. Compute $\mathrm{Cov}(X,Y)$. **[4 points]**

Solution

Since X, Y are indicators, $\mathbb{E}[X]=0.70$, $\mathbb{E}[Y]=0.60$, and $\mathbb{E}[XY]=\mathbb{P}(X=1,Y=1)=0.40$. Thus

$$\mathrm{Cov}(X,Y)=0.40-(0.70)(0.60)=-0.02.$$

Weekly Quiz 8

Instructor Key

Sampling distributions, the LLN, and the CLT

20 minutes • 10 points • Show the essential steps.

1. A population has standard deviation 12. For samples of size 64, find the standard error of the sample mean. **[3 points]**

Solution

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{64}} = 1.5.$$

2. In one sentence each, distinguish the law of large numbers from the central limit theorem. **[3 points]**

Solution

The law of large numbers says the sample mean converges to the population mean as the sample grows. The central limit theorem says the standardized sampling distribution of the sample mean becomes approximately normal (under its usual conditions), describing the shape and scale of its fluctuations.

3. If 100 independent Bernoulli trials have success probability 0.4, give the approximate mean and standard deviation of the sample proportion. Standardize the event $\hat{p} \le 0.50$. **[4 points]**

Solution

$$\mathbb{E}[\hat{p}] = 0.4, \qquad \mathrm{SD}(\hat{p}) = \sqrt{\frac{0.4(0.6)}{100}} \approx 0.0490.$$

Thus

$$\mathbb{P}(\hat{p} \le 0.50) \approx \Phi\left(\frac{0.50 - 0.40}{0.0490}\right) = \Phi(2.04).$$

Weekly Quiz 9

Instructor Key

Estimation, confidence intervals, and tests

20 minutes • 10 points • Show the essential steps.

1. In a random sample, 330 of 600 respondents answer yes. Compute the usual approximate 95% confidence interval for the population proportion, using $z^* = 1.96$. **[4 points]**

Solution

$\hat{p} = 330/600 = 0.55$ and

$$\text{SE} = \sqrt{\frac{0.55(0.45)}{600}} \approx 0.02031.$$

Thus $0.55 \pm 1.96(0.02031) = 0.55 \pm 0.0398$, or approximately $(0.510, 0.590)$.

2. Give a correct repeated-sampling interpretation of a 95% confidence procedure. **[3 points]**

Solution

If we repeatedly sampled by the same design and built an interval by this same procedure, about 95% of those intervals would contain the fixed population parameter. The 95% describes the procedure's long-run coverage, not a random probability assigned to the parameter after this interval is observed.

3. A valid test returns $p = 0.03$. State an appropriate conclusion at level $\alpha = 0.05$, and name one conclusion the p-value does *not* justify. **[3 points]**

Solution

Since $0.03 < 0.05$, reject the null hypothesis at the 5% level; the data are relatively incompatible with the null model. The result does not say that the null hypothesis has probability 0.03, nor does it by itself establish a large or practically important effect.

Weekly Quiz 10

Instructor Key

Bayesian inference and prediction

20 minutes • 10 points • Show the essential steps.

1. A Bernoulli success probability has prior $p \sim \text{Beta}(2, 2)$. After observing 8 successes and 2 failures, find the posterior distribution. **[4 points]**

   Solution

   Beta–Bernoulli conjugacy adds successes to the first shape and failures to the second. Hence

   $$p \mid \text{data} \sim \text{Beta}(2 + 8, 2 + 2) = \text{Beta}(10, 4).$$

2. Find the posterior mean and the posterior predictive probability that the next trial is a success. **[3 points]**

   Solution

   For $\text{Beta}(10, 4)$ the posterior mean is $10/(10 + 4) = 5/7$. The posterior predictive success probability is also $\mathbb{E}[p \mid \text{data}] = 5/7$.

3. Contrast a 95% Bayesian credible interval with a 95% frequentist confidence interval in one or two sentences. **[3 points]**

   Solution

   Given the prior and model, a 95% credible interval contains 95% of the posterior probability for the parameter. A 95% confidence interval comes from a procedure that covers the fixed parameter in 95% of repeated samples; it does not assign posterior probability without additional assumptions.